Arista Access Node JAY-ACC-47 QoS/Congestion Event Resolution SOP for Critical Alarm ALARM_7913

Node: JAY-ACC-47 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7913

Description

The occurrence of QoS/congestion events on Arista Access Node JAY-ACC-47 at the Jayanagar Metro Site is primarily attributable to Microbursts, interface tail drops, and QoS queue starvation, which are exacerbated by inadequate buffer sizing, insufficient policing or shaping, or misconfigured QoS policies. This phenomenon is further complicated by the role-specific characteristics of Access nodes, which prioritize low-latency and high-bandwidth forwarding, necessitating the optimization of buffer allocation, queue scheduling, and QoS policing to mitigate congestion and ensure reliable data transmission. The Arista EOS (Extensible Operating System) enables fine-grained control over QoS policies, allowing for the precise calibration of buffer sizes, policing rates, and queue scheduling algorithms to optimize network performance and prevent congestion-related issues.

Pre-requisite Checks

- Verify the site location and node ID to ensure the correct node is being addressed.

- Confirm the presence of redundant links and verify their operational status.

- Ensure adequate power availability and no signs of overheating or equipment malfunction.

- Validate the network configuration and verify that no other nodes or links are impacted by the QoS/congestion event.

Potential Root Causes

Cause	Probability
Inadequate buffer sizing or incorrect allocation of buffer resources.	High
Insufficient policing or shaping of high-bandwidth flows.	Medium
Misconfigured QoS policies or incorrect queue scheduling.	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos statistics interface Ethernet1/1, Expected Output: Input packets: 1234567, Output packets: 7654321, Packet drops: 2345678, Description: This command displays the interface statistics for Ethernet1/1, providing insights into packet drops, input/output packets, and other key metrics to help diagnose congestion-related issues.
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```
Command: show qos policy all, Expected Output: Policy Name | Match | Action | Bandwidth | Priority, Description: This command lists all QoS policies, showcasing their match criteria, actions, and associated bandwidth and priority settings, enabling the identification of potential policy misconfigurations.

Command: show hardware buffer allocation, Expected Output: Buffer Allocation: 1234567, Queue Depth: 7654321, Buffer Size: 2345678, Description: This command displays the current buffer allocation, queue depth, and buffer size for the node, providing essential information to diagnose and troubleshoot buffer-related issues.
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Resolution Steps

Step 1: Step: Verify and adjust buffer sizing and allocation, Command: config hardware buffer allocation , Description: Adjust the buffer size and queue depth to ensure optimal allocation and prevent congestion-related issues.

Step 2: Step: Reconfigure QoS policies to optimize policing and shaping, Command: config qos policy policing shaping , Description: Reconfigure QoS policies to ensure adequate policing and shaping of high-bandwidth flows, preventing congestion and ensuring reliable data transmission.

Step 3: Step: Verify and adjust queue scheduling algorithms, Command: config qos policy queue scheduling , Description: Verify and adjust queue scheduling algorithms to optimize data transmission and prevent congestion-related issues.

Post-Check Verification

```
Command: show qos statistics interface Ethernet1/1, Expected Output: Input packets: 1234567, Output packets: 7654321, Packet drops: 0, Description: Verify that packet drops have been eliminated and interface statistics are within acceptable ranges.

Command: show qos policy all, Expected Output: Policy Name | Match | Action | Bandwidth | Priority, Description: Verify that QoS policies are correctly configured and prioritized.

Command: show hardware buffer allocation, Expected Output: Buffer Allocation: 1234567, Queue Depth: 7654321, Buffer Size: 2345678, Description: Verify that buffer allocation and queue depth are optimized and within acceptable ranges.
```

Roll-back Procedures

- Step: Restore original QoS policies, Command: config qos policy restore, Description: Rollback to the original QoS policy configuration to prevent further issues.

- Step: Revert buffer sizing and allocation, Command: config hardware buffer allocation , Description: Revert the buffer size and queue depth to their original settings to prevent further congestion-related issues.